

# Callie Clark

[callieclark@nyu.edu](mailto:callieclark@nyu.edu) | (831) 227-9754 | Brooklyn, NY | [callieclark.net](http://callieclark.net)

---

NYU PhD candidate focused on large-scale mobility modeling, network science, and spatial data science. Designs end-to-end pipelines to translate human movement data into actionable insights for infrastructure, economic and community development, and public policy.

---

## EDUCATION

**New York University** **Dec 2026 (expected)**

PhD, Urban Systems (GPA: 3.85), Civil and Urban Engineering Department

Relevant Coursework: Urban Planning, Causal Inference, Machine Learning & Optimization, Complex Urban Systems

**University of California, Berkeley** **May 2021**

Master of Science, Systems Engineering (GPA: 3.96), Civil and Environmental Engineering Department

Relevant Coursework: Data Science, Scalable Spatial Analytics, Behavior Modeling, Energy Systems and Control

**University of California, Berkeley** **May 2019**

Bachelor of Science, Civil and Environmental Engineering (GPA: 3.73)

Relevant Coursework: Computer Programming, Data Science, Systems Analysis, Cyber-Physical Systems

## SKILLS

**Programming:** Python (advanced; Pandas, GeoPandas, scikit-mobility, NetworkX, OSMnx), R

**Machine Learning:** Feature engineering, model evaluation, supervised learning

**Data & Systems:** Large-scale mobility data, geospatial pipelines, graph analytics, longitudinal data

**Methods:** Causal inference, optimization, complex systems, network science

## SELECTED RESEARCH & ENGINEERING EXPERIENCE

**Research Fellow - CUSP, New York University** **Aug. 2024 - Present**

- Lead end-to-end research lifecycles, including formulating novel research questions, designing methodologies, and supervising student researchers.
- Engineered high-dimensional features from city-scale individual mobility data to optimize model performance.
- Architected data pipelines integrating mobility patterns with ancillary behavioral datasets (SafeGraph POIs, Google Reviews, EV Charger locations, Census Data) to quantify large-scale shifts in human dynamics.

**Research Fellow - Marron Institute, New York University** **Jan. 2023 - Present**

- Constructed 5-year longitudinal, directed mobility networks representing U.S. neighborhood connectivity, enabling temporal analysis of population dynamics at national scale.
- Proposed and validated network metrics to establish a novel neighborhood typology based on complex mobility patterns.

**NSF Graduate Research Fellow – New York University** **Jan. 2021- Dec 2022**

- Developed a multidimensional access index to quantify NYC resource allocation needs.
- Architected a ML-driven data pipeline integrating OSMnx, GTFS, Google Distance-Matrix API, and Uber.
- Published findings, resulting in an invitation to discuss urban food policy with the NYC Mayor's Office.
- Collaborated with engineers and interdisciplinary researchers to design scalable data pipelines, using open-source methodologies.

**Consulting Research Engineer - UC Berkeley** **Jan. - Sept. 2022**

- Secured \$80,000 in funding by co-authoring a proposal focused on modeling equitable zero-emissions mobility.
- Optimized public EV charger placement by developing multi-objective metrics considering equity, economic impact, and environmental benefits.
- Managed technical project workflows, leading bi-weekly meetings and mentoring undergraduate researchers.

Associate Data Scientist - Lawrence Berkeley National Laboratory  
Student Research Assistant - Lawrence Berkeley National Laboratory

June - Dec. 2019  
Jan. 2018 - June 2019

- Built automated data pipelines in Python process real-time occupancy data for building energy optimization.
- Investigated Deep Reinforcement Learning (DRL) control algorithms to identify research gaps in autonomous building system management.

## **PUBLICATIONS (Urban Systems, Mobility, and Causal Inference)**

M. Mavin De Silva, **Callie Clark**, Tadachika Nakayama, Takahiro Yabe. Causal spillover effects of electric vehicle charging station placement on local businesses: a staggered adoption study. [Under Review: <https://arxiv.org/abs/2511.19507v1>]

**Clark, Callie**, Christa Perfit, and Alice Reznickova. "A multi-dimensional access index: Exploring emergency food assistance in New York City." [Health & Place](#) 89 (2024): 103319

**Clark, Callie**, Chitra Dangwal, Dylan Kato, and Marta Gonzalez. "A network spatial analysis simulating response time to calls for service at variable staffing levels: A case study on strategic police defunding in the city of Chicago." [The European Physical Journal Special Topics](#) (2022): 1-9.

## **FELLOWSHIPS & AWARDS**

**National Science Foundation** Graduate Research Fellow - Highly selective over 13,000 applicants **2020 - 2025**  
**New York University** Urban Doctoral Fellow- Competitive Cross-University Program **2023 -2024**  
**ITS Berkeley** Statewide Research Transportation Program Grant – Awarded \$80,000 Project Budget **2021 – 2022**  
**UC Berkeley** Engineering Dean's Office Departmental Award – Awarded \$5,000 Research Support **2018 -2019**

## **TEACHING & MENTORSHIP**

**Graduate Student Fellow – NYU Office of Undergraduate Research** **May – June 2025**

- Mentored undergraduate researchers through the Summer Undergraduate Research Incubator (SURI), guiding students from conceptual research ideas to structured, data-driven methodologies.

**Guest Lecturer – University of Colorado Boulder** **April 2025,2026**

- Delivered an invited lecture on quantitative research methods to the master’s program in Environment and Sustainability; invited to return for the 2026 academic year.

**Consulting Research Engineer – UC Berkeley** **Jan. – Sept. 2022**

- Mentored an undergraduate engineering researcher on modeling the environmental benefits of electric vehicle adoption, providing technical guidance and feedback on research design and analysis.

**Design Assistant – T-PREP, UC Berkeley** **July – Aug. 2020, 21**

- Mentored engineering transfer students on systems-level problem solving, including cyber-physical system design and executive-level technical communication.

## **SELECTED PRESENTATIONS & INVITED TALKS**

- International Research Footprint: Presented original urban dynamics and mobility research at premier global venues, including **Netmob 2025 (France)**, the **Conference on Complex Systems (Italy)**, and the **International Conference on Computational Social Science (Sweden)**.
- Specialized Domain Expertise: presented original research on neighborhood connectivity networks as part of a multi-lab session on transit access and neighborhood change at **ACSP 2025 (Minneapolis)**.
- Policy Impact & Advocacy: Invited to discuss Food Policy with the **NYC Mayor’s Office**, following presentation at the **International Conference on Urban Affairs (New York City, 2023)**.